PYTHON FOR DATA SCIENCE · LESSON 11 OF 29

Object-Oriented Python: Classes, Objects, Inheritance and Dataclasses

Model real things as objects. Learn classes, **init**, attributes and methods, encapsulation, inheritance, polymorphism, special (dunder) methods, class and static methods, properties and dataclasses, with a small data-pipeline example.

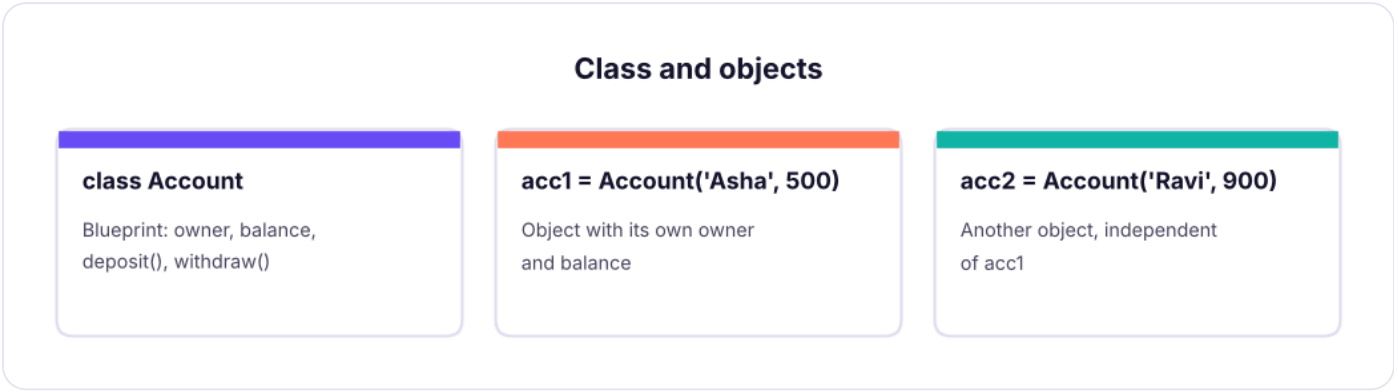
Python class 9 | Python 3.13 | Intermediate | 110 min

By the end of this lesson you can

- Define classes with attributes and methods and create objects
- Use inheritance, super() and polymorphism to share and customise behaviour
- Implement **str**, **repr**, **eq** and **len** to make objects behave like built-ins
- Use @property, @classmethod and @staticmethod appropriately
- Choose between a plain class, a dataclass and a dictionary

Why classes?

A **class** bundles **data** (attributes) and **behaviour** (methods) into one unit. An **object** is one instance of a class. Libraries you already use are classes: a pandas `DataFrame` has data (rows) and methods (`.groupby()`).



A class is the blueprint; objects are the things built from it, each with its own data.

Python

```python
class Account:
    bank = "SkillBank"                      # class attribute: shared by all objects

    def __init__(self, owner, balance=0):    # constructor: runs when you create an object
        self.owner = owner                    # instance attributes: belong to this object
        self.balance = balance

    def deposit(self, amount):
        if amount ⩽ 0: raise ValueError("amount must be positive")
        self.balance += amount
        return self.balance

    def withdraw(self, amount):
        if amount > self.balance: raise ValueError("insufficient funds")
        self.balance -= amount
        return self.balance

a, b = Account("Asha", 500), Account("Ravi", 900)
a.deposit(250); b.withdraw(100)
print(a.owner, a.balance, b.owner, b.balance, Account.bank, a.bank)
try: a.withdraw(5000)
except ValueError as e: print("Error:", e)
```

OUTPUT

```
Asha 750 Ravi 800 SkillBank SkillBank
Error: insufficient funds
```

`self` is the object the method was called on. Python passes it automatically: `a.deposit(250)` is `Account.deposit(a, 250)`.

Special (dunder) methods

Methods with double underscores customise how objects behave with `print`, `==`, `len`, `+` and more.

Python

```python
class Account:
    def __init__(self, owner, balance=0): self.owner, self.balance = owner, balance
    def __repr__(self): return f"Account({self.owner!r}, {self.balance})"      # for developers
    def __str__(self): return f"{self.owner}: ₹{self.balance:,}"               # for print()
    def __eq__(self, other): return (self.owner, self.balance) == (other.owner, other.balance)
    def __lt__(self, other): return self.balance < other.balance               # enables sorting
    def __add__(self, other): return Account(self.owner + "&" + other.owner, self.balance + other.balance)

x, y = Account("Asha", 5000), Account("Ravi", 9000)
print(x, "|", repr(x), "|", x == Account("Asha", 5000), "|", sorted([y, x]), "|", x + y)
```

OUTPUT

```
Asha: ₹5,000 | Account('Asha', 5000) | True | [Account('Asha', 5000), Account('Ravi', 9000)] | Asha&Ravi: ₹14,000
```

Method	Used by
`__init__`	Creating the object
`__repr__` , `__str__`	`repr()` , `print()`
`__eq__` , `__lt__`	`==` , `<` , sorting
`__len__` , `__getitem__`	`len(obj)` , `obj[i]`
`__iter__` , `__next__`	`for` loops
`__enter__` , `__exit__`	`with` blocks
`__call__`	Calling the object like a function

Encapsulation and properties

Python has no strict private members; a leading underscore (`_balance`) means "internal, please do not touch". A **property** exposes an attribute while keeping control over it.

Python

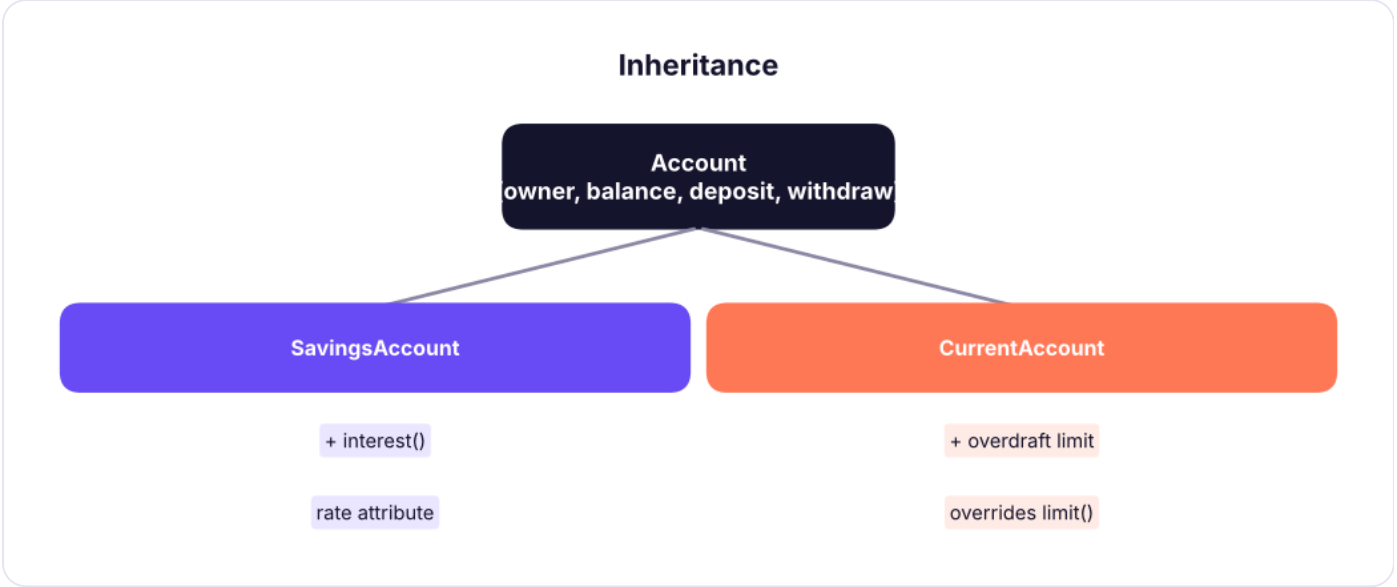
```python
class Temperature:
    def __init__(self, celsius): self._c = celsius
    @property
    def celsius(self): return self._c
    @celsius.setter
    def celsius(self, v):
        if v < -273.15: raise ValueError("below absolute zero")
        self._c = v
    @property
    def fahrenheit(self): return self._c * 9 / 5 + 32          # computed, read-only
t = Temperature(25)
t.celsius = 30
print(t.celsius, t.fahrenheit)
try: t.celsius = -500
except ValueError as e: print("Error:", e)
```

OUTPUT

```
30 86.0
Error: below absolute zero
```

Inheritance and polymorphism

A **child** class reuses a **parent** and adds or overrides behaviour. `super()` calls the parent's version.



Savings and Current accounts inherit everything from Account and add their own rules.

Python

```
class Account:
    def __init__(self, owner, balance=0): self.owner, self.balance = owner, balance
    def withdraw(self, amount):
        if amount > self.limit(): raise ValueError("limit exceeded")
        self.balance -= amount
    def limit(self): return self.balance
    def __str__(self): return f"{type(self).__name__}({self.owner}, {self.balance})"

class Savings(Account):
    def __init__(self, owner, balance=0, rate=0.04):
        super().__init__(owner, balance)            # reuse the parent's setup
        self.rate = rate
    def interest(self): return round(self.balance * self.rate, 2)

class Current(Account):
    def __init__(self, owner, balance=0, overdraft=10000):
        super().__init__(owner, balance); self.overdraft = overdraft
    def limit(self): return self.balance + self.overdraft          # override

accounts = [Savings("Asha", 50000), Current("Ravi", 2000)]
for acc in accounts:                                               # polymorphism: same
call, different behaviour
    acc.withdraw(5000)
    print(acc, "| limit now", acc.limit())
print(accounts[0].interest(), isinstance(accounts[0], Account), issubclass(Current,
Account), [c.__name__ for c in Current.__mro__])
```

OUTPUT

```
Savings(Asha, 45000) | limit now 45000
Current(Ravi, -3000) | limit now 7000
1800.0 True True ['Current', 'Account', 'object']
```

Polymorphism means different classes respond to the same method name in their own way, so code like the loop above works with any account type.

> **NOTE** **Composition often beats inheritance**
>
> "Is-a" relationships (a Savings **is an** Account) suit inheritance. "Has-a" relationships (a Pipeline **has** steps) suit composition: store other objects as attributes. Prefer composition when in doubt.

Class and static methods

Python

```
class Order:
    GST = 0.18
    def __init__(self, amount): self.amount = amount
    @classmethod
    def from_string(cls, text):                    # alternative constructor: "₹1,200"
        return cls(float(text.replace("₹", "").replace(",", "")))
    @staticmethod
    def is_valid(amount): return amount > 0        # no self/cls: just a function in the
class namespace
    def total(self): return round(self.amount * (1 + self.GST), 2)
o = Order.from_string("₹1,200")
print(o.amount, o.total(), Order.is_valid(-5))
```

OUTPUT

```
1200.0 1416.0 False
```

Dataclasses: classes for data

For classes that mainly store data, `@dataclass` generates `__init__`, `__repr__` and `__eq__` for you.

Python

```
from dataclasses import dataclass, field, asdict

@dataclass(order=True)
class Employee:
    name: str
    dept: str
    salary: float = 0.0
    skills: list = field(default_factory=list)       # mutable default done safely
    def raise_salary(self, pct): self.salary = round(self.salary * (1 + pct / 100), 2)

staff = [Employee("Kavya", "HR", 97000, ["Excel"]), Employee("Rohan", "IT", 120000),
Employee("Asha", "HR", 88000)]
staff[0].raise_salary(10)
print(staff[0], "|", staff[0] == Employee("Kavya", "HR", 106700.0, ["Excel"]), "|",
max(staff, key=lambda e: e.salary).name)
print(asdict(staff[1]), sorted(staff)[0].name)
import pandas as pd
print(pd.DataFrame([asdict(e) for e in staff]).groupby("dept")["salary"].mean())
```

OUTPUT

```
Employee(name='Kavya', dept='HR', salary=106700.0, skills=['Excel']) | True | Rohan
{'name': 'Rohan', 'dept': 'IT', 'salary': 120000, 'skills': []} Asha
dept
HR     97350.0
IT    120000.0
Name: salary, dtype: float64
```

A small data-pipeline class

Python

```python
import pandas as pd

class CleaningPipeline:
    """Holds a list of cleaning steps and applies them in order."""
    def __init__(self): self.steps = []
    def add(self, name, func): self.steps.append((name, func)); return self      # return
self allows chaining
    def run(self, df):
        for name, func in self.steps:
            before = len(df); df = func(df)
            print(f"  {name:<18} rows {before} → {len(df)}")
        return df

raw = pd.DataFrame({"city": [" Pune", "pune ", "Delhi", None, "Delhi"], "sales": [100, 100,
250, 80, -5]})
pipe = (CleaningPipeline()
        .add("drop missing city", lambda d: d.dropna(subset=["city"]))
        .add("tidy text", lambda d: d.assign(city=d.city.str.strip().str.title()))
        .add("remove negatives", lambda d: d[d.sales ⩾ 0])
        .add("drop duplicates", lambda d: d.drop_duplicates()))
print(pipe.run(raw))
```

OUTPUT

```
  drop missing city  rows 5 → 4
  tidy text          rows 4 → 4
  remove negatives   rows 4 → 3
  drop duplicates    rows 3 → 2
    city  sales
0   Pune    100
2  Delhi    250
```

This is the idea behind scikit-learn pipelines (ML track): objects with `fit` / `transform` methods that you chain.

When to use what

Need	Use
A few related values	Tuple / dict
Record with named fields, no logic	`@dataclass` or namedtuple
Data plus behaviour, state that changes	Class
Share behaviour across related types	Inheritance (or composition)
One function	Just a function

Common mistakes

Mistake	Fix
Forgetting `self` in the method signature	`def method(self, ...)`
Forgetting `self.` when reading an attribute	`self.balance`, not `balance`
Mutable class attribute shared by all objects (`items = []`)	Create it in `__init__`
Overusing inheritance	Prefer composition and small functions
Missing `super().__init__()` in the child	Call it to initialise parent state
Comparing objects without `__eq__`	Compares identity, not content; use a dataclass

Summary

KEY TAKEAWAYS

- A class is a blueprint; `__init__` sets up each object; `self` refers to the current object.
- Dunder methods (`__str__` , `__repr__` , `__eq__` , `__lt__` , `__len__`) make objects behave like built-ins.
- Inheritance reuses and overrides parent behaviour with `super()` ; polymorphism lets one call work across types.
- `@property` guards attributes; `@classmethod` builds alternative constructors; `@staticmethod` groups helpers.
- `@dataclass` removes boilerplate for data holders. Prefer composition over deep inheritance.

Quick quiz

6 questions · tap an answer to check it

1. What does `self` refer to?

A The class

B The specific object the method is called on

C The module

D The parent class

Answer: B. Python passes the instance automatically.

2. Which method runs when you create an object?

A `__str__`

B `__init__`

C `__new__` only

D `__call__`

Answer: B. `__init__` initialises the new instance.

3. What does `super().__init__(owner, balance)` do in a child class?

A Deletes the parent

B Runs the parent's constructor to set up inherited attributes

C Creates a new class

D Nothing

Answer: B. It reuses the parent's initialisation.

4. Which decorator gives a read-only computed attribute?

A `@staticmethod`

B `@property`

C `@classmethod`

D `@cache`

Answer: B. Access without parentheses.

5. What does `@dataclass` generate automatically?

A Only `__str__`

B `__init__` , `__repr__` and `__eq__`

C A database table

D Nothing

Answer: B. It removes boilerplate for data holders.

6. What is polymorphism?

A Having many parents

B Different classes responding to the same method name in their own way

C Copying a class

D Hiding attributes

Answer: B. The loop `for acc in accounts: acc.withdraw()` works for any account type.

EXERCISE **OOP problems**

(1) Create a `Rectangle` class with `area()` and `perimeter()`, and a `Square` subclass. (2) Add `__str__` and `__eq__` to a `Point(x, y)` class. (3) Build a `Stack` class (push, pop, peek, `__len__`). (4) Make a dataclass `Product` with price and `with_gst()`, then sort a list of products by price. (5) Write a `Counter`-like class that counts calls using a class attribute.

Show solution (worked answer)

Python

```
class Rectangle:
    def __init__(self, w, h): self.w, self.h = w, h
    def area(self): return self.w * self.h
    def perimeter(self): return 2 * (self.w + self.h)
class Square(Rectangle):
    def __init__(self, side): super().__init__(side, side)
print(Rectangle(3, 4).area(), Square(5).perimeter())
class Point:
    def __init__(self, x, y): self.x, self.y = x, y
    def __str__(self): return f"({self.x}, {self.y})"
    def __eq__(self, o): return (self.x, self.y) == (o.x, o.y)
print(Point(1, 2), Point(1, 2) == Point(1, 2))
class Stack:
    def __init__(self): self._d = []
    def push(self, x): self._d.append(x)
    def pop(self): return self._d.pop()
    def peek(self): return self._d[-1]
    def __len__(self): return len(self._d)
st = Stack(); st.push(1); st.push(2); print(st.pop(), len(st))
from dataclasses import dataclass
@dataclass(order=True)
class Product:
    price: float
    def with_gst(self): return round(self.price * 1.18, 2)
print(sorted([Product(500), Product(120)]), Product(100).with_gst())
```

INTERVIEW CORNER

Questions you may be asked

Q: Class attribute vs instance attribute? A class attribute is shared by all instances (defined in the class body); an instance attribute belongs to one object (set in `__init__`).

Q: What is the difference between `__str__` and `__repr__`? `__str__` is the readable form for users (`print`); `__repr__` is the unambiguous form for developers, ideally something that could recreate the object.

Q: Explain the four OOP pillars. Encapsulation (bundle and protect data), inheritance (reuse), polymorphism (same interface, different behaviour), abstraction (expose what, hide how).

Q: Inheritance or composition? Use inheritance for a true is-a relationship; use composition (objects holding other objects) for flexibility and lower coupling.

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.